F79MB Statistical Models B
Weekly study plan 2025-26

[See also "Course induction AY25-26", "Welcome to the Course", and "Modules" with course content that will be released weekly on Canvas.]

At start of course: **Install R and RStudio** on your computer (if not already done).
You can download R and R Studio using below links:
Download and Install R: https://www.r-project.org/
Download and Install free version of R Studio Desktop: https://posit.co/download/rstudio-desktop/

	Notes	**Computer practical work**
Week 1	Chapter 1. Review of estimation and testing Chapter 2-1. Quantile function; Empirical quantile function Chapter 2-2. (i) QQ plots (ii) Refinement	Sheet 1 [Normal QQ plots]
Week 2	Chapter 2-3. Checking data modelled by exponential distribution Chapter 3-1. (i) Chi-square gof tests (ii) Using R to perform χ^2- test for E.g. 2.1 Chapter 3-2. Kolmogorov-Smirnov gof tests	Sheet 2 [Exponential QQ plots, normal QQ plots of transformations of data]
Week 3	Chapter 4-1. Non-parametric estimation Chapter 4-2. (i) Bootstrap estimation (ii) Bootstrap estimation for C.I Chapter 4-3. Hypothesis tests using bootstrap	Sheet 3 [Chisq goodness of fit test, Kolmogorov-Smirnov test]
Week 4	Chapter 5-1. Regression – brief revision Chapter 5-2. Regression illustration using R Chapter 5-3(i). Model diagnostics: residuals and Influence Chapter 5-3(ii). Model diagnostics: residuals and Influence	Sheet 4 [Bootstrap] **Release of Project 1 (Thursday, February 5)**
Week 5	Chapter 5-3(iii). Model diagnostics: residuals and influence Chapter 5-4. Correlation	Sheet 5 [Simple linear regression]
Week 6	**Consolidation week**	
Week 7	Chapter 6-1. Multiple regression: parameter estimation Chapter 6-2. Multiple regression: example Chapter 6-3. Multiple regression: hypothesis tests and confidence intervals Chapter 6-4. Multiple regression: interpretation Chapter 6-5. Multiple regression: example 6.4	Sheet 6 [Multiple linear regression] **Submission deadline for Project 1 (Monday, 23rd February, by 3:30 PM local campus time)** **Individual feedback is expected to be released by Monday, 16th March 2026.**
Week 8	Chapter 6-6. Multiple regression: example 6.5 Chapter 7. Other regression techniques Chapter 8-1. Generalised linear models: introduction Chapter 8-2. Generalised linear models: definition of GLM Chapter 8-3. Fitting a GLM	Sheet 7 [Generalised linear models]
Week 9	Chapter 8-4. Model checking: Analysis of deviance and residuals for GLMs Chapter 8-5. Model comparison: Analysis of deviance and Akaike Information Criterion	Sheet 8 [Generalised linear models]
Week 10	Chapter 9. Discrete and categorical data analysis Chapter 10. Categorical data analysis: two-way classifications.	Sheet 9 [Contingency tables] **Release of Project 2 (Thursday, March 19)**
Week 11	Revision work or free to work on assessment Practical Lab: Computer Practical 9	

Week 12	**Free to work on assessment**	
Week 13 (Study Week)		**Submission deadline for Project 2 (Wednesday, 8th April, by 3:30 PM local campus time)** **Individual feedback is expected to be released by Friday, 8th May 2026.**